Chapter 5
Stratified Random Sampling

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Description

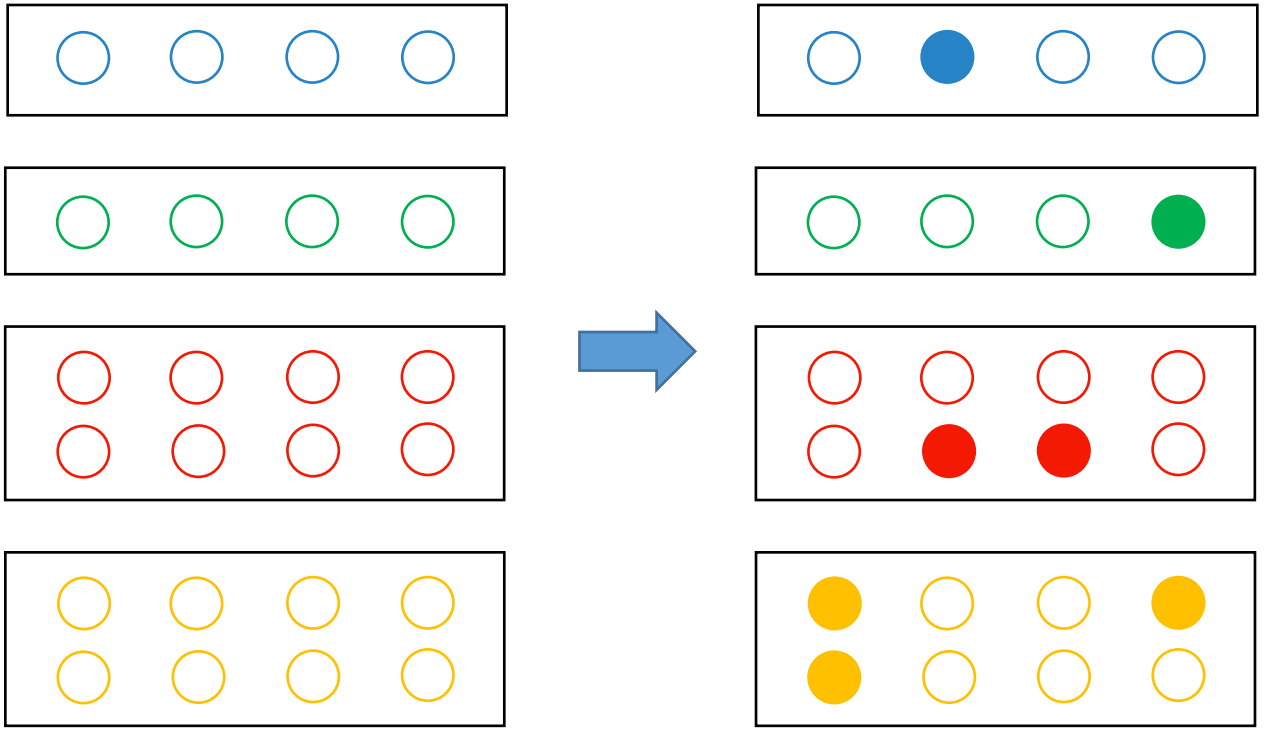
The population of N units: divided into L subpopulations

- These subpopulations (or strata) are non-overlapping, and together they comprise the whole of the population.
- N_h : the population size of the hth stratum, must be known

$$N_1 + N_2 + \cdots N_L = N$$

- Within each stratum, samples are drawn independently across the strata.
- n_h : the sample size in the hth stratum
- Stratified random sampling: perform SRS in each stratum

In each of the L strata, choose n_h units from N_h units without replacement, and independently across strata.



Why stratification?

- Control for domains of study

  Data of known precision are wanted for certain subdivisions.

- Administrative convenience

- Flexibility in design and estimation

  Sampling problems may differ markedly in different parts of the population.

- Efficiency

  If each stratum is homogeneous, a precise estimate of any stratum mean can be obtained from a small sample in that stratum.

Notation

- h: the stratum
- i: the unit within the stratum
- The following symbols all refer to stratum h

y_{hi} value obtained for the ith unit

$W_h = \frac{N_h}{N}$ stratum weight

$f_h = \frac{n_h}{N_h}$ sampling fraction in the stratum

$\bar{Y}_h = \frac{1}{N_h} \sum_{i=1}^{N_h} y_{hi}$ true mean

$\bar{y}_h = \frac{1}{n_h} \sum_{i=1}^{n_h} y_{hi}$ sample mean

$$Y_h = N_h \bar{Y}_h = \sum_{i=1}^{N_h} y_{hi}$$ true total

$$y_h = n_h \bar{y}_h = \sum_{i=1}^{n_h} y_{hi}$$ sample total

$$S_h^2 = \frac{1}{N_h - 1} \sum_{i=1}^{N_h} (y_{hi} - \bar{Y}_h)^2$$ true variance

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i=1}^{n_h} (y_{hi} - \bar{y}_h)^2$$ sample variance

Properties of the estimates

For the population mean per unit, the estimate is

$$\bar{y}_{st} = \sum_{h=1}^{L} W_h \bar{y}_h = \frac{\sum_{h=1}^{L} N_h \bar{y}_h}{N}$$

The sample mean

$$\bar{y} = \frac{\sum_{h=1}^{L} n_h \bar{y}_h}{n}$$

- Difference: Weight!
- $\bar{y}_{st} = \bar{y}$, when $\frac{n_h}{n} = \frac{N_h}{N}$ or $\frac{n_h}{N_h} = \frac{n}{N}$ or $f_h = f$
- proportional allocation of n_h , a self-weighting sample

Theorem 5.1 If in every stratum the sample estimate $\bar{y}_h$ is unbiased, then $\bar{y}_{st}$ is an unbiased estimate of the population mean $\bar{Y}$.

Theorem 5.2 If the samples are drawn independently in different strata,

$$V(\bar{y}_{st}) = \sum_{h=1}^{L} W_h^2 V(\bar{y}_h)$$

where $V(\bar{y}_h)$ is the variance of $\bar{y}_h$ over repeated samples from stratum h.

- If $\bar{y}_h$ is unbiased, and sample selection is independent in different strata, then $\bar{y}_{st}$ is an unbiased estimate with variance $\sum W_h^2 V(\bar{y}_h)$.
- The variance of $\bar{y}_{st}$ depends only on the variances of the estimates of the individual stratum means $\bar{Y}_h$!

- Principle: divide a highly variable population into strata such that all items have close values within a stratum
- Reason: the use of the correct stratum weights N_h / N in estimation

Theorem 5.3 For stratified random sampling, the variance of the estimate $\bar{y}_{st}$ is

$$V(\bar{y}_{st}) = \sum_{h=1}^{L} W_h^2 V(\bar{y}_h) = \sum_{h=1}^{L} W_h^2 \frac{1-f_h}{n_h} S_h^2$$

Corollary 1. If the sampling fractions n_h / N_h are negligible in all strata, is

$$V(\bar{y}_{st}) = \sum_{h=1}^{L} W_h^2 V(\bar{y}_h) = \sum_{h=1}^{L} \frac{W_h^2}{n_h} S_h^2 = \frac{1}{N^2} \sum_{h=1}^{L} \frac{N_h^2 S_h^2}{n_h}$$

Corollary 2. With proportional allocation, we use $n_h = nN_h / N$, then the variance reduces to

$$V(\bar{y}_{st}) = \frac{1-f}{n}\sum_{h=1}^{L} W_h S_h^2 = \sum_{h=1}^{L} \frac{N_h}{N}\frac{S_h^2}{n}\left(\frac{N-n}{N}\right)$$

Corollary 3. If sampling is proportional and the variances in all strata have the same value S_w^2, we obtain the simplest result

$$V(\bar{y}_{st}) = \frac{1-f}{n} S_w^2$$

Theorem 5.4 If $\hat{Y}_{st} = N\bar{y}_{st}$ is the estimate of the population total Y, then

$$V(\hat{Y}_{\mathrm{st}}) = N^2 V(\bar{y}_{\mathrm{st}}) = \sum_{h=1}^{L} N_h(N_h - n_h)\frac{S_h^2}{n_h}$$



Example

Table 5.1 shows the 1920 and 1930 number of inhabitants, in thousands, of 64 large cities in the United States. The cities are arranged in two strata, the first containing the 16 largest cities and the second the remaining 48 cities.

The total number of inhabitants in all 64 cities in 1930 is to be estimated from a sample of size 24. Find the standard error of the estimated total for

1. a simple random sample,
2. a stratified random sample with proportional allocation,
3. a stratified random sample with 12 units drawn from each stratum.

SIZES OF 64 CITIES (IN 1000'S) IN 1920 AND 1930

1920 Size (x_{hi})				1930 Size (y_{hi})			
Stratum $h = 1$	2			Stratum 1	2		
797	314	172	121	900	364	209	113
773	298	172	120	822	317	183	115
748	296	163	119	781	328	163	123
734	258	162	118	805	302	253	154
588	256	161	118	670	288	232	140
577	243	159	116	1238	291	260	119
507	238	153	116	573	253	201	130
507	237	144	113	634	291	147	127
457	235	138	113	578	308	292	100
438	235	138	110	487	272	164	107
415	216	138	110	442	284	143	114
401	208	138	108	451	255	169	111
387	201	136	106	459	270	139	163
381	192	132	104	464	214	170	116
324	180	130	101	400	195	150	122
315	179	126	100	366	260	143	134

Note. Cities are arranged in the same order in both years.

Totals and sums of squares

	1920		1930	
Stratum	$\sum(x_{hi})$	$\sum(x_{hi}^2)$	$\sum(y_{hi})$	$\sum(y_{hi}^2)$
1	8,349	4,756,619	10,070	7,145,450
2	7,941	1,474,871	9,498	2,141,720

For the complete population in 1930, $Y = 19568,\ S^2 = 52448$

1. For simple random sampling,

$$V(\hat{Y}_{srs}) = \frac{N^2 S^2}{n}(1-f) = \frac{64^2 \times 52448}{24}\left(1-\frac{24}{64}\right) = 5594453$$

$$\sigma(\hat{Y}_{srs}) = 2365$$

2. In proportional allocation, we have $n_1 = 6,\ n_2 = 18$.

   For individual strata, the variances are

$$S_1^2 = \frac{7145450 - 10070^2/16}{16-1} = 53843,\ S_2^2 = 5581$$

Note that the stratum with the largest cities has a variance nearly 10 times that of the other stratum. $S_1^2 = 53843$ $S_2^2 = 5581$

$$V(\hat{Y}_{prop}) = N^2 \frac{1-f}{n} \sum_{h=1}^{2} W_h S_h^2 = \frac{N-n}{n} \sum_{h=1}^{2} N_h S_h^2$$

$$= \frac{40}{24}\left(16 \times 53843 + 48 \times 5581\right) = 1882293$$

$$\sigma(\hat{Y}_{prop}) = 1372$$

3. For $n_1 = n_2 = 12$,

$$V(\hat{Y}_{equal}) = \sum_{h=1}^{2} N_h (N_h - n_n) \frac{S_h^2}{n_n} = \frac{16 \times (16-12) \times 53843}{12} + \frac{48 \times (48-12) \times 5581}{12}$$

$$= 1090827$$

$$\sigma(\hat{Y}_{equal}) = 1044$$

The estimated variance and confidence limits

If a simple random sample is taken within each stratum, an unbiased estimate of S_h^2 is

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i=1}^{n_h} (y_{hi} - \bar{y}_h)^2$$

Theorem 5.5 With stratified random sampling, an unbiased estimate of the variance of $\bar{y}_{st}$ is

$$v(\bar{y}_{st}) = s^2(\bar{y}_{st}) = \sum_{h=1}^{L} W_h^2 \frac{1 - f_h}{n_h} s_h^2 = \frac{1}{N^2} \sum_{h=1}^{L} N_h (N_h - n_h) \frac{s_h^2}{n_h}$$

An alternative form:

$$s^2(\bar{y}_{st}) = \sum_{h=1}^{L} \frac{W_h^2 s_h^2}{n_h} - \sum_{h=1}^{L} \frac{W_h s_h^2}{N}$$

The second term represents the reduction due to fpc.

The confidence limits:

Population mean $\bar{y}_{st} \pm t s(\bar{y}_{st})$

Population total $N\bar{y}_{st} \pm t N s(\bar{y}_{st})$

These formulas assume that $\bar{y}_{st}$ is normally distributed.



Example

In order to investigate average number of persons in a household in a district, the households are classified into two strata, urban and rural. A stratified random sample with 10 households were drawn from each stratum. The results are listed in the following table.

We want to estimate the average number of persons in a household as well as a 95% confidence limits.

Stratum	Total number of households	Number of persons in a household									
		1	2	3	4	5	6	7	8	9	10
1	250	3	2	3	4	3	3	4	5	2	3
2	500	3	4	5	5	4	3	6	2	4	4

h	n_h	N_h	W_h	f	$\bar{y}_h$	$W_h\bar{y}_h$	s_h^2
1	10	250	0.3333	0.04	3.2	1.0667	0.8444
2	10	500	0.6667	0.02	4.0	2.6667	1.3333
Total	20	750				3.7333	

$$\bar{y}_{\text{st}} = \sum_{h=1}^{2} W_h \bar{y}_h \approx 3.733$$

$$v(\bar{y}_{\text{st}}) = \sum_{h=1}^{2} W_h^2 (1 - f_h) \frac{s_h^2}{n_h} \approx 0.06708$$

$$\sqrt{v(\bar{y}_{\text{st}})} \approx 0.259$$

$$\bar{y}_{\text{st}} \pm t\sqrt{v(\bar{y}_{\text{st}})} \quad \Rightarrow \quad (3.24,\ 4.24)$$

$$t = 1.96$$

Optimum allocation

Choose the sample sizes n_h in the strata

- to minimize $V(\bar{y}_{st})$ for a specified cost of taking the sample
- to minimize the cost for a specified value of $V(\bar{y}_{st})$

The simplest cost function is

$$C = c_0 + \sum_{h=1}^{L} c_h n_h$$

where c_0 represents an overhead cost.

- This cost function is appropriate when the major item of cost is that of taking the measurements on each unit.

Theorem 5.6 In stratified random sampling with a linear cost function, the variance of the estimated mean $\bar{y}_{st}$ is a minimum for a specified cost C, and the cost is a minimum for a specified variance $V(\bar{y}_{st})$, when n_h is proportional to $W_h S_h / \sqrt{c_h}$.

Proof

$$C = c_0 + \sum_{h=1}^{L} c_h n_h$$

$$V = V(\bar{y}_{st}) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{n_h}(1 - f_h) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{n_h} - \frac{1}{N}\sum_{h=1}^{L} W_h S_h^2$$

Choosing the n_h to minimize V for fixed C or C for fixed V are both equivalent to minimizing the product.

$$V'C' = \left(V + \frac{1}{N} \sum W_h S_h^2 \right) \left(C - c_0 \right) = \left(\sum \frac{W_h^2 S_h^2}{n_h} \right) \left(\sum c_h n_h \right)$$

Cauchy-Schwarz inequality:

If a_h, b_h are two sets of L positive numbers,

$$\left(\sum a_h^2 \right) \left(\sum b_h^2 \right) \geq \left(\sum a_h b_h \right)^2$$

Equality occurring if and only if b_h / a_h is constant for all h.

- Optimum allocation

$$\frac{n_h}{n} = \frac{\frac{W_h S_h}{\sqrt{c_h}}}{\sum_{h=1}^{L} \frac{W_h S_h}{\sqrt{c_h}}} = \frac{\frac{N_h S_h}{\sqrt{c_h}}}{\sum_{h=1}^{L} \frac{N_h S_h}{\sqrt{c_h}}}, \quad h = 1,2,\cdots,L$$

- In a given stratum, take a larger sample if
  - the stratum is larger;
  - the stratum is more variable internally;
  - sampling is cheaper in the stratum .

- The corresponding variance:

$$V_{opt}(\bar{y}_{st}) = \frac{\sum_{h=1}^{L} W_h S_h / \sqrt{c_h}}{n} \sum_{h=1}^{L} W_h S_h \sqrt{c_h} - \frac{1}{N} \sum_{h=1}^{L} W_h S_h^2$$

- The value of n

  If cost C is fixed,

$$n = \frac{(C - c_0)\sum(W_h S_h / \sqrt{c_h})}{\sum(W_h S_h \sqrt{c_h})} = \frac{(C - c_0)\sum(N_h S_h / \sqrt{c_h})}{\sum(N_h S_h \sqrt{c_h})}$$

  If V is fixed,

$$n = \frac{\sum(W_h S_h \sqrt{c_h})\sum(W_h S_h / \sqrt{c_h})}{V + (1/N)\sum W_h S_h^2}$$

- An important special case: $c_h = c$
- Optimum allocation for fixed cost reduces to optimum allocation for fixed sample size, since $C = c_0 + cn$.
- Neyman allocation

Theorem 5.7 (Neyman allocation) In stratified random sampling, $V(\bar{y}_{st})$ is minimized for a fixed total size of sample n if

$$n_h = n\frac{W_h S_h}{\sum_{h=1}^{L} W_h S_h} = n\frac{N_h S_h}{\sum_{h=1}^{L} N_h S_h}, \quad h = 1,2,\cdots,L$$

- The minimum variance with fixed n

$$V_{\min}(\bar{y}_{st}) = \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2 - \frac{1}{N}\sum_{h=1}^{L} W_h S_h^2$$

The second term represents the reduction due to fpc.



Example

Suppose we are interested in the household income in a city, which consists of two districts denoted by District-1 and District-2. Let n=500. It is known that there are 20000 households in District-1, and 50000 households in District-2; the standard errors of the household income are $S_1 = 2500$ dollars and $S_2 = 2000$ dollars for District-1 and District-2 respectively. Further assume that the ratio of the costs for investigating one household in the two districts is 2:3.

Determine the values of n_h for proportional, optimum, and Neyman allocations.

h	N_h	W_h	S_h	c_h	W_hS_h	$W_hS_h/\sqrt{c_h}$
1	20000	0.2857	2500	2	714.2857	505.0763
2	50000	0.7143	2000	3	1428.5714	824.7861
Total	70000	1.0000	—	—	2142.8571	1329.8624

1.Proportional

$$n_1 = nW_1 = 500 \times 0.2857 = 142.86 \approx 143$$

$$n_2 = nW_2 = 500 \times 0.7143 = 357.14 \approx 357$$

2.Nayman

$$n_1 = n.\frac{W_1S_1}{\sum_{h=1}^{2} W_hS_h} = 500 \times \frac{714.2857}{2142.8571} = 166.67 \approx 167$$

$$n_2 = n.\frac{W_2S_2}{\sum_{h=1}^{2} W_hS_h} = 500 \times \frac{1428.571}{2142.857} = 333.33 \approx 333$$

h	N_h	W_h	S_h	c_h	$W_h S_h$	$W_h S_h / \sqrt{c_h}$
1	20000	0.2857	2500	2	714.2857	505.0763
2	50000	0.7143	2000	3	1428.5714	824.7861
Total	70000	1.0000	—	—	2142.8571	1329.8624

3.Optimum

$$n_1 = n\frac{W_1 S_1 / \sqrt{c_1}}{\sum_{h=1}^{2}(W_h S_h / \sqrt{c_h})} = 500 \times \frac{505.0763}{1329.8624} = 189.898 \approx 190$$

$$n_2 = n\frac{W_2 S_2 / \sqrt{c_2}}{\sum_{h=1}^{2}(W_h S_h / \sqrt{c_h})} = 500 \times \frac{824.7861}{1329.8624} = 310.102 \approx 310$$

Relative precision of stratified random and simple random sampling

- Stratification nearly always results in a smaller variance for the estimated mean or total than is given by a comparable simple random sample.
- However, it is not true that any stratified random sample gives a smaller variance in a simple random sample.
- If the values of the n_h are far from optimum, stratified sample may have a higher variance.

- For the estimated means: $V_{ran}, V_{prop}, V_{opt}$

Theorem 5.8 If terms in $1/N_h$ are ignored relative to unity,

$$V_{opt} \le V_{prop} \le V_{ran}$$

where optimum allocation is for fixed n, i.e. $n_h \propto N_h S_h$.

Simple random sampling $V_{ran} = \dfrac{1-f}{n} S^2$

Proportional stratification $V_{prop} = \dfrac{1-f}{n} \sum_{h=1}^{L} W_h S_h^2 = \dfrac{1}{n} \sum_{h=1}^{L} W_h S_h^2 - \dfrac{1}{N} \sum_{h=1}^{L} W_h S_h^2$

Stratification with Neyman $V_{opt} = \dfrac{1}{n} \left(\sum_{h=1}^{L} W_h S_h \right)^2 - \dfrac{1}{N} \sum_{h=1}^{L} W_h S_h^2$

From the standard algebraic identity for the analysis of variance of a stratified population,

$$S^2 = \sum_{h=1}^{L} \frac{N_h - 1}{N - 1} S_h^2 + \sum_{h=1}^{L} \frac{N_h}{N - 1} (\bar{Y}_h - \bar{Y})^2$$

And

$$\frac{N_h}{N - 1} \approx \frac{N_h - 1}{N - 1} \approx \frac{N_h}{N} = W_h$$

Hence $$S^2 \approx \sum_{h=1}^{L} W_h S_h^2 + \sum_{h=1}^{L} W_h (\bar{Y}_h - \bar{Y})^2$$

$$V_{ran} = \frac{1-f}{n} S^2 \approx \frac{1-f}{n} \sum_{h=1}^{L} W_h S_h^2 + \frac{1-f}{n} \sum_{h=1}^{L} W_h (\bar{Y}_h - \bar{Y})^2$$

$$= V_{prop} + \frac{1-f}{n} \sum_{h=1}^{L} W_h (\bar{Y}_h - \bar{Y})^2 \geq V_{prop}$$

By the definition of $V_{\min}$, we must have $V_{prop} \geq V_{opt}$

$$V_{prop} - V_{opt} = \frac{1}{n} \left[\sum_{h=1}^{L} W_h S_h^2 - \left(\sum_{h=1}^{L} W_h S_h \right)^2 \right] = \frac{1}{n} \sum_{h=1}^{L} W_h (S_h - \bar{S})^2$$

where $\bar{S} = \sum_{h=1}^{L} W_h S_h$.

We have

$$V_{ran} = V_{opt} + \frac{1}{n}\sum_{h=1}^{L} W_h (S_h - \bar{S})^2 + \frac{1-f}{n}\sum_{h=1}^{L} W_h (\bar{Y}_h - \bar{Y})^2$$

- Two components of the decrease in variance from SRS to optimum allocation:
  - the elimination of differences among the stratum means
  - the elimination of the effect of differences among the stratum standard deviations
- The second component represents the difference in variance between optimum and proportional allocation.

Proportional v.s. optimum stratification

- Optimum stratification is better when
  - The population consists of large and small institutions, stratified by some measure of size.
  - Some strata are much more expensive to sample than others.
- Proportional is preferred as
  - The optima in the allocation problem are rather flat.
  - The errors in the estimated S_h^2 would reduce the superiority of the optimum.
  - The simplicity and the self-weighting feature of proportional allocation are probably worth a 10 to 20% increase in variance.

When does stratification produce large gains in precision?

- The ideal variate for stratification is the value of y itself
  - no overlap between strata, and the variance within strata would be much smaller than the over-all variance.
- In practice, we consider three conditions
  - The population is composed of institutions varying widely in size.
  - The principal variables to be measured are closely related to the sizes of the institutions.
  - A good measure of size is available for setting up the strata.

- If the size of the institution remains stable through time, then its best practical measure is usually the size of the institution on some recent occasion when a census was taken.

TABLE 5.2

CALCULATION OF THE OPTIMUM ALLOCATION

		1920 Data			1930 Data		
Stratum	N_h	S_h	N_hS_h	n_h	S_h	N_hS_h	n_h
1	16	163.30	2612.80	11.56	232.04	3712.64	12.21
2	48	58.55	2810.40	12.44	74.71	3586.08	11.79
Totals	64		5423.20	24.00		7298.72	24.00

- The possibility of a change in the levels of the S_h^2 should always be considered when using past data.

- Geographic stratification is common—often for administrative convenience or because separate data are wanted for each stratum.
- The gains are generally modest.

TABLE 5.3

RELATIVE PRECISION OF DIFFERENT KINDS OF GEOGRAPHIC STRATIFICATION (IN PER CENT)

State	No. of Items	Stratum			
		Township	County	Type of Farming Area	State
Iowa, 1938	18	115	100	96	91
Iowa, 1939	19	121	100	97	91
Florida, 1942					
Citrus fruit area	14	144	100	..	
Truck farming area	15	111	100	..	
California, 1942	17	113	100	97	

Allocation requiring more than 100% sampling

- Occur: when the over-all sampling fraction is substantial and some strata are much more variable than others.
- If the original allocation gives $n_1 > N_1$ when there are more than two strata, the optimum revised allocation is:

$$\tilde{n}_1 = N_1; \quad \tilde{n}_h = (n - N_1)\frac{W_h S_h}{\sum_2^L W_h S_h}, \quad (h \geq 2)$$

provided that $\tilde{n}_h \leq N_h, h \geq 2.$

- If it should happen that $\tilde{n}_2 > N_2$, continue the process until all $\tilde{n}_h \leq N_h$.



Example

Suppose a population is divided into 4 strata. The values of N_h and S_h are listed in the following table. If $n=100$, determine the values of n_h if Neyman allocation is adopted.

h	N_h	S_h	N_hS_h	$\frac{N_hS_h}{\sum N_hS_h}$	n_h
1	6	500	3000	0.0833	8.3
2	20	600	12000	0.3333	33.3
3	300	20	6000	0.1667	16.7
4	500	30	15000	0.4167	41.7
Total	826		36000	1.0000	100

$$\tilde{n}_1 = 6,\ \tilde{n}_2 = (100-6)\times\frac{12000}{33000} = 34.2 > 20$$

$$\tilde{n}_1' = 6,\ \ \tilde{n}_2' = 20,\ \ \tilde{n}_3' = 21,\ \ \tilde{n}_4' = 53$$

$$V_{opt} = \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2 - \frac{1}{N}\sum_{h=1}^{L} W_h S_h^2$$

- The general formula of the variance should be used.

$$V(\bar{y}_{st}) = \sum_{h=1}^{L} W_h^2 V(\bar{y}_h) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{n_h}(1 - f_h) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{n_h} - \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{N_h}$$

- The variance for the Neyman allocation no longer holds.
- An alternative correct formula is

$$V_{\min}(\bar{y}_{st}) = \frac{(\Sigma' W_h S_h)^2}{n'} - \frac{\Sigma' W_h S_h^2}{N}$$

where n' is the revised total sample size in these strata, and Σ' denotes the summation over the strata in which $\tilde{n}_h \le N_h$.

Estimation of sample size with continuous data

- Assume: the estimate has a specified variance V.
- Some relationships

  Absolute error: $d = t \cdot SE(\hat{\theta}) = t\sqrt{V(\hat{\theta})}$

  Relative error: $r = t \cdot \frac{SE(\hat{\theta})}{\theta} = t \cdot CV(\hat{\theta})$

  Coefficient of variation: $CV(\hat{\theta}) = SE(\hat{\theta}) / \theta$

  $\Rightarrow$ $V = (d / t)^2$

  $V = (r\theta / t)^2$

- Let s_h be the estimate of S_h
- $n_h = nw_h$

- Estimation of the population mean $\bar{Y}$

$$V = \sum_{h=1}^{L} W_h^2 \frac{1-f_h}{n_h} s_h^2 = \sum_{h=1}^{L} \frac{W_h^2}{n_h} s_h^2 - \sum_{h=1}^{L} \frac{W_h^2}{N_h} s_h^2$$

$$= \frac{1}{n} \sum_{h=1}^{L} \frac{W_h^2}{w_h} s_h^2 - \frac{1}{N} \sum_{h=1}^{L} W_h s_h^2$$

- A general formula for n:

$$n = \frac{\sum \frac{W_h^2}{w_h} s_h^2}{V + \frac{\sum W_h s_h^2}{N}}$$

or

$$n_0 = \frac{1}{V} \sum \frac{W_h^2 s_h^2}{w_h}$$

$$n = \frac{n_0}{1 + \frac{1}{NV} \sum W_h s_h^2}$$

- Neyman allocation (optimum allocation for fixed n): $w_h = \frac{W_h s_h}{\sum W_h s_h}$

$$n = \frac{\left(\sum W_h s_h\right)^2}{V + \frac{\sum W_h s_h^2}{N}}$$

- Proportional allocation: $w_h = W_h = N_h / N$

$$n = \frac{\sum \frac{W_h^2}{w_h} s_h^2}{V + \frac{\sum W_h s_h^2}{N}} = \frac{\sum W_h s_h^2}{V + \frac{\sum W_h s_h^2}{N}} \quad \text{or} \quad n_0 = \frac{\sum W_h s_h^2}{V}$$

$$n = \frac{n_0}{1 + \frac{n_0}{N}}$$

- Estimation of the population total Y

$$n = \frac{\sum \frac{N_h^2}{w_h} s_h^2}{V + \sum N_h s_h^2}$$

- Neyman allocation (optimum allocation for fixed n):

$$n = \frac{\left(\sum N_h s_h\right)^2}{V + \sum N_h s_h^2}$$

- Proportional allocation:

$$n_0 = \frac{N}{V} \sum N_h s_h^2 \qquad n = \frac{n_0}{1 + \frac{n_0}{N}}$$



Example

The population consists of 196 teachers' colleges and normal schools. These were arranged in seven strata, of which one small stratum will be ignored. The first five strata were constructed by size of institution: the sixth contained colleges for women only. Estimates s_h of the S_h were computed from results for the 1943-1944 academic year. An "optimum" stratification based on these s_h was employed.

The objective was a coefficient of variation of 5% in the estimated total enrollment.

In 1943 the total enrollment for this group of colleges was 56,472.

$$0.05 \times 56472 = 2824 \qquad V = 2824^2 = 7974976$$

TABLE 5.4

DATA FOR ESTIMATING SAMPLE SIZE

Stratum	N_h	s_h	$N_h s_h$	n_h
1	13	325	4,225	9
2	18	190	3,420	7
3	26	189	4,914	10
4	42	82	3,444	7
5	73	86	6,278	13
6	24	190	4,560	10
Totals	196		26,841	56

With $V = 2824^2 = 7974976$

$$n = \frac{\left(\sum N_h s_h\right)^2}{V + \sum N_h s_h^2} = \frac{26841^2}{7974976 + 4640387} = 57.1$$



Example

In order to investigate average number of persons in a household in a district, the households are classified into two strata, urban and rural. A stratified random sample with 10 households were drawn from each stratum. The results are listed as follows.

Stratum	Number of households	Number of persons in a household									
		1	2	3	4	5	6	7	8	9	10
Urban	250	3	2	3	4	3	3	4	5	2	3
Rural	500	3	4	5	5	4	3	6	2	4	4

(1) Estimate the average number of persons in a household as well as a 95% CI.

(2) Consider the estimation of the average number of persons in a household. With a relative error 10% and a confidence level 95%, compute the total sample size and the sample sizes in the respective strata if the Neyman allocation is adopted.

(1)

h	N_h	n_h	f_h	W_h	$\bar{y}_h$	S_h^2	$W_h\bar{y}_h$
1	250	10	0.04	0.33333	3.2	0.8444	1.0667
2	500	10	0.02	0.66667	4	1.3333	2.6667

$$\bar{y}_{st} = W_1\bar{y}_1 + W_2\bar{y}_2 = 3.733$$

$$v(\bar{y}_{st}) = \sum W_h^2 \frac{1-f_h}{n_h} s_h^2 \approx 0.06708 \qquad se(\bar{y}_{st}) = 0.259$$

The 95% CI: $(\bar{y}_{st} - 1.96se, \bar{y}_{st} + 1.96se) = (3.24, 4.24)$

(2) Neyman allocation

$$V = (r\bar{y}_{st}/t)^2 = \left(\frac{0.1\times 3.733}{1.96}\right)^2 = 0.03628$$

$$n = \frac{\left(\sum W_h s_h\right)^2}{V + \frac{\sum W_h s_h^2}{N}} = 30.61 \approx 31$$

$$n_1 = n \times \frac{W_1 s_1}{\sum_{h=1}^{2} W_h s_h} = 9, \qquad n_2 = n - n_1 = 22$$

Stratified sampling for proportions

- The **ideal** stratification: place in the first stratum every unit that falls in *C*, and in the second every unit that does not.
- Principle: construct strata such that the proportion in class *C* varies as much as possible from stratum to stratum.
- The proportions of units in *C* in the *h*th stratum and in the sample from that stratum

$$P_h = \frac{A_h}{N_h} \qquad p_h = \frac{a_h}{n_h}$$

- The estimate for the proportion in the whole population:

$$p_{st} = \sum_{h=1}^{L} W_h p_h$$

Theorem 5.9 With stratified random sampling, the variance of p_{st} is

$$V(p_{st}) = \sum_{h=1}^{L} W_h^2 V(p_h) = \sum_{h=1}^{L} W_h^2 \frac{N_h - n_h}{N_h - 1} \frac{P_h Q_h}{n_h}$$

Corollary 1. With proportional allocation,

$$V(p_{st}) = \frac{1-f}{n} \sum_{h=1}^{L} W_h S_h^2 = \frac{1-f}{n} \sum_{h=1}^{L} W_h \frac{N_h P_h Q_h}{N_h - 1}$$

$$\approx \frac{1-f}{n} \sum_{h=1}^{L} W_h P_h Q_h$$

Theorem 5.10 With stratified random sampling, the estimate of the variance of p_{st} is

$$v(p_{st}) = \sum_{h=1}^{L} W_h^2 v(p_h) = \sum_{h=1}^{L} W_h^2 \frac{N_h - n_h}{N_h} \frac{p_h q_h}{n_h - 1}$$

Corollary 2. With proportional allocation,

$$v(p_{st}) = \frac{1-f}{n} \sum_{h=1}^{L} W_h s_h^2 = \frac{1-f}{n} \sum_{h=1}^{L} W_h \frac{n_h p_h q_h}{n_h - 1}$$

- The best choice of the n_h in order to minimize $V(p_{st})$
- Minimum variance for fixed total sample size

$$n_h \propto N_h \sqrt{N_h/(N_h-1)}\sqrt{P_h Q_h} \approx N_h \sqrt{P_h Q_h}$$

$$n_h \approx n \frac{N_h \sqrt{P_h Q_h}}{\sum N_h \sqrt{P_h Q_h}}$$

- Minimum variance for fixed cost: $Cost = c_0 + \sum c_h n_h$

$$n_h \approx n \frac{N_h \sqrt{P_h Q_h / c_h}}{\sum N_h \sqrt{P_h Q_h / c_h}}$$

Estimation of sample size with proportions

Let V be the desired variance in the estimate of the proportion P for the whole population

- Proportional allocation:

$$n_0 = \frac{\sum W_h s_h^2}{V} \quad \Rightarrow \quad n_0 = \frac{\sum W_h p_h q_h}{V}, \; n = \frac{n_0}{1 + n_0 / N}$$

- Neyman allocation (optimum allocation for fixed n):

$$n = \frac{\left(\sum W_h s_h\right)^2}{V + \frac{\sum W_h s_h^2}{N}} \quad \Rightarrow \quad n = \frac{\left(\sum W_h \sqrt{p_h q_h}\right)^2}{V + \frac{\sum W_h p_h q_h}{N}}$$

- Let V be the desired variance in the estimate of the total number in the population in class C, that is NP.
- Proportional allocation:

$$n_0 = \frac{N\sum N_h p_h q_h}{V},\ n = \frac{n_0}{1+n_0/N}$$

- Neyman allocation (optimum allocation for fixed n):

$$n = \frac{(\sum N_h \sqrt{p_h q_h})^2}{V + \sum N_h p_h q_h}$$